

Mechanical Displacement: A Model for Quantum Tunneling

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1. The Problem: The Identity Trap in the QGP

In conventional models of quantum tunneling, a particle overcomes an energetic barrier that it could not pass according to classical mechanics. Within the previous logic of *Andrew's Ground*, a problem emerged: if a particle (as a condensed 3D fragment) leaves the 3D skin and enters the 4D Quark-Gluon Plasma (QGP), it should theoretically dissolve into the medium and lose its individuality. Thus, "tunneling" would be impossible, as the particle would no longer exist as the same object at the end of the barrier.

2. The Solution: The Incompressible Displacement Model

In this framework, tunneling is not understood as a particle traveling through a wall, but as an **instantaneous mechanical exchange process** within an existing hardware connection (pressure column).

A. The "Filled Pipe" Analogy

Imagine a pressure column within the QGP as a rigid pipe, completely filled with an incompressible fluid (the primordial medium). At both ends of the pipe sits a valve (the 3D boundary layer of our universe).

- The pipe is already "charged." It exists as a standing connection between two points in space.
- If an additional particle (energy packet) is pressed into the pipe at **Point A** with high pressure, the incompressible medium cannot yield.

B. Instantaneous Ejection

Due to the mechanical rigidity of the QGP medium, the momentum is conducted through the pipe without any loss of time (instantaneously). Since the same pressure arrives at **Point B**, exactly the same amount of medium (an identical particle) is pushed out of the 3D skin.

3. Physical Conclusion

Mechanically speaking, the "tunneled" particle is not the exact same object that entered at Point A. However, it is an **identical copy**, as it consists of the same primordial substance (QGP) and carries the precisely transmitted quantum information (momentum, spin, energy) of the pressure column.

- **Entanglement:** A weak impulse across the column (signal transmission).
- **Tunneling:** A massive displacement across the column (matter exchange).

Tunneling is therefore not a "magical" disappearance, but a **hydraulic remote action**. The barrier in our 3D world is not traversed, but simply "bypassed" via the 4D hardware of the background.

Andrew's Ground: Where the quantum world stops playing dice and starts working.